International Journal of
Radiation Oncology
biology • physics

www.redjournal.org

Clinical Investigation

Stereotactic Radiosurgery and Fractionated Stereotactic Radiation Therapy for the Treatment of Uveal Melanoma

Gozde Yazici, MD,* Hayyam Kiratli, MD,† Gokhan Ozyigit, MD,* Sezin Yuce Sari, MD,* Mustafa Cengiz, MD,* Bercin Tarlan, MD,‡ Burce Ozgen Mocan, MD,§ and Faruk Zorlu, MD*

*Departments of *Radiation Oncology, †Ophthalmology, and §Radiology, Faculty of Medicine, Hacettepe University, Ankara, Turkey; and ‡Bascom Palmer Eye Institute, Miami, Florida*

Received Jun 3, 2016, and in revised form Feb 5, 2017. Accepted for publication Feb 10, 2017.

Summary

The results of stereotactic radiosurgery and fractionated stereotactic radiation therapy in uveal melanoma were retrospectively evaluated. It was found that higher tumor control rates are achieved with doses of ≥45 Gy in 3 fractions. The 2-year enucleation-free survival rate was 73% for all tumors and 69% for large tumors.

Purpose: To evaluate treatment results of stereotactic radiosurgery or fractionated stereotactic radiation therapy (SRS/FSRT) for uveal melanoma.

Methods and Materials: We retrospectively evaluated 181 patients with 182 uveal melanomas receiving SRS/FSRT between 2007 and 2013. Treatment was administered with CyberKnife.

Results: According to Collaborative Ocular Melanoma Study criteria, tumor size was small in 1%, medium in 49.5%, and large in 49.5% of the patients. Seventy-one tumors received <45 Gy, and 111 received ≥45 Gy. Median follow-up time was 24 months. Complete and partial response was observed in 8 and 104 eyes, respectively. The rate of 5-year overall survival was 98%, disease-free survival 57%, local recurrence-free survival 73%, distant metastasis-free survival 69%, and enucleation-free survival 73%. There was a significant correlation between tumor size and disease-free survival, SRS/FSRT dose and enucleation-free survival; and both were prognostic for local recurrence-free survival. Enucleation was performed in 41 eyes owing to progression in 26 and complications in 11.

Conclusions: The radiation therapy dose is of great importance for local control and eye retention; the best treatment outcome was achieved using ≥45 Gy in 3 fractions.

Reprint requests to: Faruk Zorlu, MD, Department of Radiation Oncology, Faculty of Medicine, Hacettepe University, Ankara, Turkey 06100. Tel: (+90) 312-3052900; E-mail: fzorlu@hacettepe.edu.tr

Conflict of interest: none.

Int J Radiation Oncol Biol Phys, Vol. 98, No. 1, pp. 152−158, 2017
0360-3016/$ - see front matter
http://dx.doi.org/10.1016/j.ijrobp.2017.02.017

Introduction

Uveal melanoma is the most common primary intraocular malignancy in adults. The goal of treatment is to preserve the eye with useful vision and prevent distant metastasis (DM) (1). Tumor size and maximum basal diameter are the most important prognostic factors for the development of DM (2).

The Collaborative Ocular Melanoma Study (COMS) (3) defined uveal melanomas, according to thickness and basal diameter, respectively, as follows: large >10 mm and ≥16 mm; medium 2.5 to 10 mm and <16 mm; small 1 to 3 mm and 5 to 16 mm. The 5-year survival rate was similar for medium-sized melanomas treated with enucleation and plaque brachytherapy (PBT). A rapid shift to eye-preserving treatment modalities occurred as local recurrence (LR) after PBT was associated with poor prognosis. Currently various radiation therapy (RT) techniques constitute the treatment of choice (4). Enucleation is now reserved for eyes unsalvageable after other treatments, or in cases in which excessive morbidity is expected without the prospect of useful vision.

The CyberKnife (CK) (Accuray, Sunnyvale, California) is a highly developed linear accelerator that performs stereotactic radiosurgery (SRS) and fractionated stereotactic RT (FSRT). Higher doses of radiation are delivered to the target compared with conventional RT, whereas the dose decreases sharply beyond the tumor, facilitating better preservation of organs at risk (OARs). Accordingly, the present study aimed to determine whether any prognostic factor could predict the outcome in uveal melanoma after SRS/FSRT.

Methods and Materials

Patients

The records of 181 patients with 182 uveal melanomas treated using SRS/FSRT between 2007 and 2013 were retrospectively reviewed. Uveal melanoma was diagnosed via indirect ophthalmoscopy. Ophthalmologic evaluation included assessment of best-corrected visual acuity (VA), anterior segment and fundus examinations, and fundus fluorescein angiography. Tumor thickness was measured via A- and B-mode ocular ultrasonography. All patients underwent pretreatment staging based on anamnesis, physical examination, blood biochemistry, and abdominal CT and/or positron emission tomography/CT. Patients with DM at the time of diagnosis were excluded. The study protocol was approved by the institutional ethics committee and was performed in accordance with the Declaration of Helsinki.

SRS and FSRT

All patients underwent SRS/FSRT via the CK in the supine position. A thermoplastic mask was used for immobilization. Peribulbar injection of 2% lidocaine (5 mL) was administered 20 to 30 minutes before planning CT and magnetic resonance imaging (MRI) and every treatment fraction. Slice thickness was 1 mm for planning CT and MRI.

Target volume and OARs were delineated using the fused CT/MRI data set. Gross tumor volume (GTV) was defined as the contrast-enhanced lesion. Clinical target volume was equal to GTV, and planning target volume was defined as clinical target volume plus a 1-mm margin in all directions, excluding the side of the fovea. The OARs (eg, fellow eye, lenses, optic nerves, and the optic chiasm) were also contoured. The fovea was not delineated because it is not possible to be visualized via routine CT and MRI. Treatment plans were optimized so that ≥99% of the planning target volume received 95% of the prescribed dose. Six-D skull tracking was used during treatment.

Tumor response was assessed after 12 weeks via ophthalmologic examination, B-mode ocular ultrasonography, and MRI. Patients were examined every 3 months during the first 2 years, every 6 months during the next 3 years, and annually thereafter. Follow-up data were obtained from department charts, hospital notes, referring doctors, and the General Directorate of Population and Citizenship Affairs.

Statistical analysis

Statistical analysis was performed using PASW Statistics for Windows version 18.0 (SPSS, Chicago, IL). Primary endpoints were local recurrence-free survival (LRFS) and enucleation-free survival (EFS). Secondary endpoints included overall survival (OS), disease-free survival (DFS), distant metastasis-free survival (DMFS), VA, and late treatment toxicity. Local control (LC) was defined as the lack of tumor progression (ie, an increase in tumor volume). Complete response was defined as the disappearance of the tumor, and partial response as a >50% decrease in the tumor volume. Other responses were accepted stable disease. Survival analysis was performed using the Kaplan-Meier method, and the results were compared using the log−rank test. All time-related events were calculated from the last day of treatment to the last follow-up visit or death.

The Cox proportional hazards model was used for multivariate analysis. Gender (male vs female), tumor size (large vs small/medium), and treatment dose (<45 Gy vs ≥45 Gy) were included as potentially significant covariates after univariate analysis. Factors with a significant contribution to survival estimation ($P<.20$) were preserved in the final multivariate model. The hazard ratio with a 95% confidence interval was calculated.

Late toxicity was defined as toxicity that developed >3 months after the end of treatment. The incidence was calculated as the total number of patients who developed toxicity at any time divided by the total number of assessable patients. The level of statistical significant was set at $P<.05$.

Results

Patient and treatment characteristics

In total, 181 patients were retrospectively reviewed. Median patient age was 54 years (range, 18-82 years). Among the patients, 104 (58%) were male, and 77 (42%) were female. Median tumor diameter and thickness was 10 mm (range, 2-12 mm) and 8 mm (range, 1.5-18 mm), respectively. According to COMS classification, 1% of the tumors were small, 49.5% were medium, and 49.5% were large (3). According to TNM classification, 23 patients (13%) had T1, 36 (20%) had T2, 70 (38%) had T3, and 14 (8%) had T4 tumors. Tumor stage in 39 patients could not be determined. All T1 tumors were small/medium, and all T4 tumors were large, according to COMS. One T2 tumor was large and 35 were small/medium, whereas 54 of T3 tumors were large and 16 were small/medium.

Age, gender, and treatment dose were similar in patients with large and small/medium tumors ($P=.9$, $P=.2$, and $P=.9$, respectively). All patients were treated with their eyes closed, either owing to ptosis via anesthesia or manually with a patch. In total, 71 tumors (39%) received <45 Gy to the GTV, versus ≥45 Gy in 111 (61%). In all, 61% of the large tumors and 59% of the small/medium tumors received ≥45 Gy ($P=.5$). Treatment details are shown in Table 1. Median follow-up time was 24 months (range, 2-79 months) in all patients and did not differ according to tumor size.

Table 1 Treatment details

Characteristic	Median (range)
Total dose (Gy)	54 (10-66)
Fraction dose (Gy)	18 (10-22)
No. of fractions	3 (1-3)
Treatment time (min)	42 (35-50)
BED3	226.8 (26-330)
CI	1.39 (1.04-3.66)
HI	1.18 (1.09-1.49)
Collimator size (mm)	5 (5-20)
Normalization (%)	85 (67-92)
No. of beams	167 (58-284)
Size of the GTV (mm^3)	561 (78-9158)
Maximum dose of the GTV (Gy)	62.65 (11.5-89.55)
Maximum dose of the ipsilateral eye (Gy)	60.67 (5.05-89.55)
Maximum dose of the contralateral eye (Gy)	4.12 (0.83-13.43)
Maximum dose of the ipsilateral lens (Gy)	16.26 (1.92-77.92)
Maximum dose of the contralateral lens (Gy)	1.75 (0.15-8.07)
Maximum dose of the ipsilateral optic nerve (Gy)	19.58 (1.86-74.26)
Maximum dose of the contralateral optic nerve (Gy)	2.4 (0.14-13.68)
Maximum dose of the optic chiasm (Gy)	1.66 (0.23-9.21)

Abbreviations: BED = biologically effective dose; CI = conformity index; GTV = gross tumor volume; HI = homogeneity index.

Local recurrence-free survival

The 2-year LRFS rate was 82%. The tumor response was total in 8 tumors (4%), partial in 104 (57%), stable in 25 (14%), and progressed in 26 (14%). The final response was unknown in 19 tumors. The overall LC rate was 75%. All recurrences occurred in the treatment field; marginal recurrence was not observed. The treatment plan for a patient with a large tumor is given in Figure 1.

As tumor size increased and treatment dose decreased, the LRFS rate decreased significantly (Table 2). After ≥45 Gy, the 2-year LRFS rate was 77% for large tumors and 98% for small/medium tumors ($P=.009$). Male gender was associated with a lower LRFS rate (Table 2). Tumor size and treatment dose were significant factors for LRFS according to multivariate analysis (Table 3).

Overall and disease-free survival

The 2-year OS rate was 98%. One patient died because of disease recurrence, and another died of an unknown cause. Seven patients were lost to follow-up and were not included in survival analyses. No significant factor was identified for OS via univariate or multivariate analysis (Table 2).

The 2-year DFS rate was 78%. As tumor size increased and treatment dose decreased, the DFS rate decreased significantly (Table 2). After ≥45 Gy, the 2-year DFS rate was 76% for large tumors and 93% in small/medium tumors ($P=.06$). According to multivariate analysis, tumor size was the only significant factor for DFS (Table 3).

Distant metastasis-free survival

Distant metastases developed in 16 patients (9%). The 2-year DMFS rate was 90%. After ≥45 Gy, the 2-year DMFS rate was 83% for large tumors and 93% in small/medium tumors ($P=.2$). No significant factors for DMFS were identified via univariate or multivariate analysis.

Enucleation-free survival

Forty-one eyes (23%) were enucleated during follow-up. The 2-year EFS rate was 73%, and it was significantly higher in female patients and doses ≥45 Gy (Table 2). Nonetheless, tumor size did not significantly affect the EFS rate (Fig. 2, Table 2). The RT dose was the only significant factor associated with EFS according to multivariate analysis (Table 3).

The number of enucleated patients did not differ significantly according to tumor size ($P=.06$); however, significantly more patients receiving <45 Gy underwent enucleation as compared with those receiving ≥45 Gy

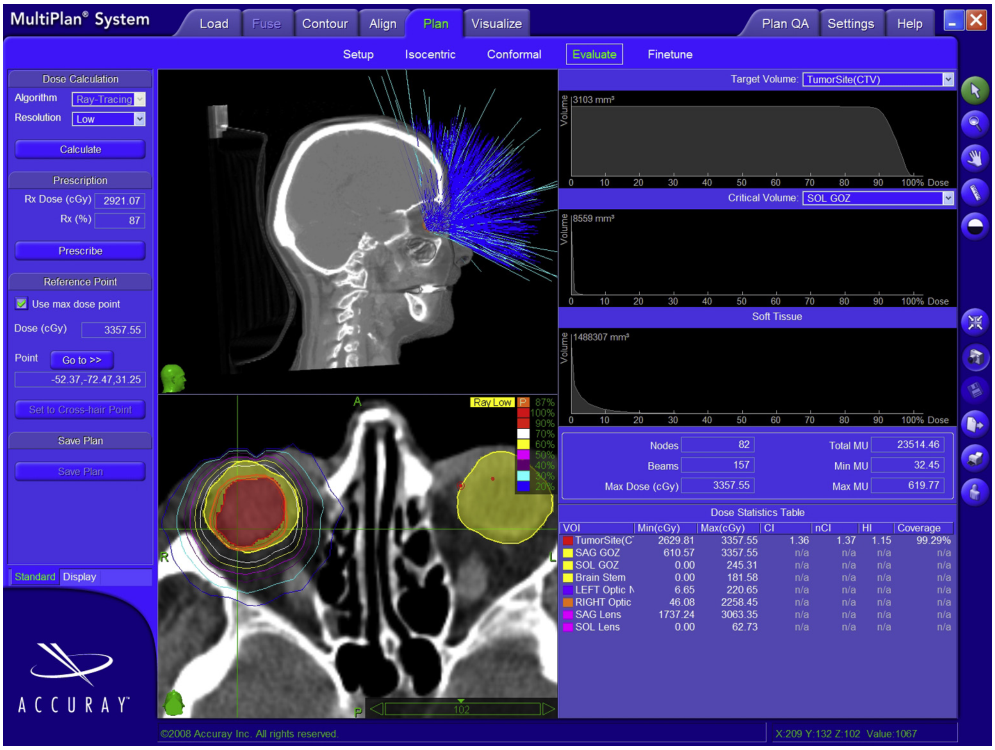

Fig. 1. CyberKnife plan for a patient with a choroidal melanoma 18 mm high. Upper left: Pathway of the beams. Lower left: The lines surrounding the gross tumor volume represent isodose levels. The treatment dose was prescribed to the 87% isodose level, represented by the orange line. The outer dark blue line represents the 20% isodose level. Upper right: Dose–volume histograms of the clinical target volume, left eye, and soft tissues. Lower right: Dose statistics. (A color version of this figure is available at www.redjournal.org.)

(n = 24 vs 17) (P = .007). Analysis of the relationship between tumor size and RT dose showed that 50% of large tumors receiving <45 Gy and 28% receiving ≥45 Gy underwent enucleation (P = .07), whereas in small/medium tumors the respective rates were 33% and 14% (P = .06). Similarly, there was not a significant difference in the enucleation rate between eyes with large and small/medium tumors receiving <45 Gy (P = .2) and ≥45 Gy (P = .1).

Enucleation was performed owing to tumor progression in 26 tumors, RT-associated complication (ie, rubeosis iridis) in 11, and nonresponse to RT in 2. Enucleation could not be specified in 2 patients because they underwent surgery at other facilities. The indication for enucleation was similar in large and small/medium tumors (P = .2). Among the enucleated eyes, 24 received <45 Gy and 17 received ≥45 Gy (P = .4). Analysis of the correlation between the

Table 2 Results of univariate analysis

Survival and factor	2-y OS (%)	P	2-y DFS (%)	P	2-y LRFS (%)	P	2-y DMFS (%)	P	2-y EFS (%)	P
Age (y)										
≤60	99	.1	80	.6	81	.6	91	.7	73	.6
>60	95		71		80		86		74	
Gender										
Male	96	.1	71	.09	75	.05	88	.6	65	.03*
Female	100		88		88		92		84	
COMS size										
Small-medium	98	.9	70	.02*	91	.009*	95	.2	83	.055
Large	98		88		73		87		70	
SRS/FSRT dose (Gy)										
<45	97	.2	69	.04*	72	.01*	92	.2	62	.002*
≥45	100		84		87		89		81	

Abbreviations: COMS = Collaborative Ocular Melanoma Study; DFS = disease-free survival; DMFS = distant metastasis-free survival; EFS = enucleation-free survival; FSRT = fractionated stereotactic radiation therapy; LRFS = local recurrence-free survival; OS = overall survival; SRS = stereotactic radiosurgery.

* Statistically significant.

Table 3 Results of multivariate analysis for LRFS, DFS, and EFS

Factor	HR	95% confidence interval	*P*
LRFS			
CK dose (Gy)			
≥45	1	1.08-4.75	.03
<45	2.27		
Tumor size*			
Other	1	1.32-6.82	.009
Large	3		
Gender			
Female	1	0.98-4.83	.057
Male	2.17		
DFS			
CK dose (Gy)			
≥45	1	0.92-3.56	.086
<45	1.8		
Tumor size*			
Other	1	1.16-4.68	.018
Large	2.33		
Gender			
Female	1	0.88-3.51	.108
Male	1.76		
EFS			
CK dose (Gy)			
≥45	1	1.3-4.56	.005
<45	2.43		
Tumor size*			
Other	1	0.94-3.31	.08
Large	1.76		
Gender			
Female	1	0.99-3.66	.053
Male	1.91		

Abbreviations: CK = CyberKnife; HR = hazard ratio. Other abbreviations as in Table 2.

* According to the Collaborative Ocular Melanoma Study.

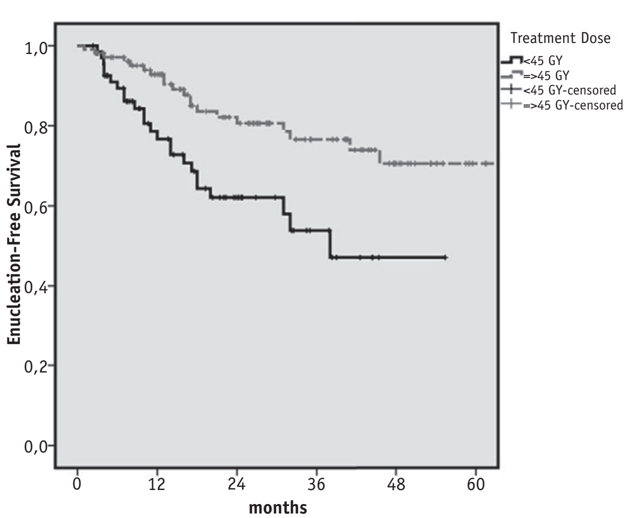

Fig. 2. Enucleation-free survival rates according to tumor size.

need for enucleation and tumor size and RT dose showed no significant difference in the enucleation rate between large and small/medium tumors ($P=.7$ and $P=.7$, respectively). Likewise, analysis of the correlation between the need for enucleation and RT dose and tumor size showed no significant difference in the enucleation rate between tumors that received <45 Gy and ≥45 Gy ($P=.3$ and $P=.4$, respectively).

Visual acuity

Visual acuity was measured in 168 eyes at the time of diagnosis; 44 eyes had 20/20 vision, 106 had partial vision, and 18 had complete loss of vision. At the last follow-up, VA was measured in 98 eyes; VA decreased in 65 eyes (66%), was stable in 32 eyes (33%), and increased in 1 eye (1%). There were not any statistically significant differences in change in VA according to tumor size or RT dose ($P=.5$ and $P=.9$, respectively). When tumor size was analyzed in terms of the effect of treatment dose on change in VA, no difference was observed between large and small/medium tumors ($P=.5$ and $P=.9$, respectively). When treatment dose was analyzed in terms of the effect of tumor size on change in VA, the rates were similar between <45 Gy and ≥45 Gy ($P=.5$ and $P=.9$, respectively).

Toxicity

Radiation retinopathy developed in 76 eyes (42%) in median 12 months, and resolved in 18 eyes in median 16 months. Median eye dose was significantly higher in the eyes with retinopathy than in those without (63 Gy vs 52 Gy, $P=.04$).

Vitreous hemorrhage was observed in 8 eyes (4%) in median 10 months but was present in half of them at the time of diagnosis. Retinal detachment was observed in 19 eyes (10%) in median 10 months but was present in 15 of them at the time of diagnosis. Radiation papillopathy developed in 20 eyes (11%) in median 17 months, neovascular glaucoma (NVG) developed in 6 eyes (3%) in median 14 months, and optic atrophy developed in 7 eyes (4%) in median 38 months. Loss of eye lashes was noted in 3 eyes (2%) at the last follow-up.

Radiation-induced cataract developed in 27 eyes (15%) in median 14 months. Median lens dose was significantly higher in the eyes with radiation-induced cataract than in those without (29 Gy vs 16 Gy, $P=.007$).

Discussion

Despite local treatment innovations, the survival rate for uveal melanoma remains unchanged (5, 6). Enucleation was the treatment of choice in the 1970s, but currently enucleation is reserved for medium tumors with local failure and for large tumors not suitable for eye preservation. The present study observed that the eye could be preserved with SRT/FSRT in the majority of patients with large tumors.

Plaque brachytherapy is a good alternative to enucleation for medium tumors (7). Survival rates after external beam RT and PBT were found to be similar (8); however, no prospective trial has compared external beam RT and PBT. Enucleation is still considered the treatment of choice for large tumors and in patients not expected to regain vision (9-11).

Another alternative to enucleation is charged particle therapy (CPT). The risk of death and enucleation after CPT and PBT was found to be similar; however, the risk of LR and rates of cataract formation and radiation retinopathy were significantly lower with CPT (12). Mishra et al (13) reported higher LC and eye preservation rates after CPT compared with PBT. Bensoussan et al (14) reported a 5-year LC rate of 94% after PBT in large tumors, with 5-year OS rates of 68% and 52% for T3 and T4 tumors, respectively. Their enucleation rate was 19.5%, primarily owing to treatment complications. Although results are satisfactory with CPT, because of its expense CPT facilities are very rare. Therefore, CPT is not a very good alternative in many countries. To the best of our knowledge no study has compared CPT and SRS/FSRT with photons.

Stereotactic RT for ocular melanoma in humans was first performed via the Gamma Knife (GK; Elekta, Stockholm, Sweden) by Zambrano et al (15). Treatment is completed in 1 day because an invasive form of immobilization is required. The prescribed dose in SRS studies varies widely, with 5-year LC, OS, and eye retention rates of 87% to 97%, 55% to 91%, and 77% to 90%, respectively (16-20).

Another approach to SRS/FSRT is the linear accelerator—based technique. With the help of the noninvasive immobilization technique, the fractionated scheme decreases the rate of side effects. Dieckmann et al (21) delivered 60 to 70 Gy in 5 fractions in tumors with a median 5.4-mm thickness. The LC, DMFS, and secondary enucleation rates were 98%, 97%, and 7.7%, respectively. They reported the rates of retinopathy, cataract, optic neuropathy, and NVG as 25.5%, 18.9%, 20%, and 8.8%, respectively. Dunavoelgyi et al (22) reported 5-year and 10-year LC and overall DM rates of 95.9%, 92.6%, and 15%, respectively, after 50 to 70 Gy in 5 fractions.

Muller et al (23) delivered 50 Gy in medium tumors and reported 2-year LC and OS rates of 100% and 67%, respectively, with no enucleation required. The update of this study reported 5-year LC and DMFS rates of 96% and 75%, respectively (24). Biewald et al (25) reported an 89% eye-retention rate after $\geq$25 Gy via GK, followed by endoresection with or without PBT. In our study the eye retention rate was 85% after $\geq$45 Gy.

A study on various SRS techniques reported that CK was better than GK for preserving OARs (26). One of the first studies on CK reported preliminary results in 5 patients not eligible for local resection or PBT (27). After 3×20 Gy, 3 tumors regressed and 2 were stable, and VA improved in 2 patients and remained stable in 3. Muacevic et al (28) delivered 18 to 22 Gy SRS via CK in 20 patients with tumors with a median 7-mm thickness. They reported no LR or secondary NVG after a mean 13 months. Choi et al (29) delivered 36 to 39 Gy in 5 fractions to 6 patients with medium tumors; 5 patients had a partial response without serious complication. Another study on CK reported a 5-year eye retention rate of 73% after a median 20-Gy of SRS (30). The enucleation rate was 12%, mostly owing to LR; however, the number of large tumors was not mentioned, and the complication rate was higher than in our study.

In our study there was a significant dose—response relationship, with higher LC and EFS rates with $\geq$45 Gy. The 2-year LRFS rate for small/medium tumors receiving $\geq$45 Gy was 98%, comparable to earlier series. Nonetheless, SRS/FSRT was observed as an alternative to immediate enucleation for large tumors. We achieved an eye retention rate of 72% after $\geq$45 Gy, with a DFS rate similar to those in earlier reports (31).

Despite the fractionated scheme, complications can occur after FSRT. Secondary enucleation might be required owing to complication or tumor progression (10). Approximately 20% of patients undergo enucleation after RT, mainly because of complications rather than tumor progression (32, 33). Neovascular glaucoma is the most common cause of secondary enucleation and occurs in $\leq$35% of patients after SRS (34). The rate of RT-induced cataract and retinopathy was 83% and 63%, respectively (35). Complication rates in our study are comparable to those in earlier reports. Although Klingenstein et al (36) reported similar quality of life after enucleation and severe functional loss after CK, they concluded that CK can be considered the treatment of choice, because it provides significantly better quality of life compared with enucleation.

The major limitation of our study is its retrospective design, which resulted in incomplete data on patient and tumor characteristics, and treatment results. Nineteen patients were lost to follow-up, and more patients were lost while evaluating VA. Median follow-up is 24 months and is too short for observing the development of RT-related toxicity. In addition, the distance to OARs was not measured. Most importantly, our study focused on the results of SRS/FSRT and did not compare it with enucleation or other RT techniques. To obtain accurate data concerning whether SRS/FSRT is noninferior to PBT-CPT, these modalities should be compared in randomized trials. The comparison of the results of SRS/FSRT versus PBT-CPT is beyond the scope of this report.

Conclusion

Standard RT techniques for uveal melanoma are PBT and PT. Although our findings show that SRS/FSRT is another good alternative to enucleation, it cannot be compared with PBT-PT, because the duration of follow-up was too short. Using SRS/FSRT, better control of large tumors was achieved with ≥45 Gy in 3 fractions. On the basis of the present findings, we recommend increasing the radiation dose, as well as limiting the maximum eye and lens dose to 50 Gy and 15 Gy, respectively, so as to increase the eye retention rate. Additional studies with longer follow-up are needed to more clearly discern the benefit of stereotactic photon therapy.

References

1. Scotto J, Fraumeni JF Jr., Lee JA. Melanomas of the eye and other noncutaneous sites: Epidemiologic aspects. *J Natl Cancer Inst* 1976; 56:489-491.
2. Shields CL, Furuta M, Thangappan A, et al. Metastasis of uveal melanoma millimeter-by-millimeter in 8033 consecutive eyes. *Arch Ophthalmol* 2009;127:989-998.
3. Margo CE. The Collaborative Ocular Melanoma Study: An overview. *Cancer Control* 2004;11:304-309.
4. Tarlan B, Kıratlı H. Uveal Melanoma: Current trends in diagnosis and management. *Turk J Ophthalmol* 2016;46:123-137. Review.
5. Strickland D, Lee JA. Melanomas of eye: Stability of rates. *Am J Epidemiol* 1981;113:700-702.
6. Damato B. Does ocular treatment of uveal melanoma influence survival? *Br J Cancer* 2010;103:285-290.
7. Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: V. Twelve-year mortality rates and prognostic factors: COMS report no. 28. *Arch Ophthalmol* 2006;124:1684-1693.
8. Abrams MJ, Gagne NL, Melhus CS, et al. Brachytherapy vs. external beam radiotherapy for choroidal melanoma: Survival and patterns-of-care analyses. *Brachytherapy* 2016;15:216-223.
9. Bell DJ, Wilson MW. Choroidal melanoma: Natural history and management options. *Cancer Control* 2004;11:296-303.
10. Damato B, Lecuona K. Conservation of eyes with choroidal melanoma by a multimodality approach to treatment: An audit of 1632 patients. *Ophthalmology* 2004;111:977-983.
11. Damato B, Heimann H. Personalized treatment of uveal melanoma. *Eye (Lond)* 2013;27:172-179.
12. Wang Z, Nabhan M, Schild SE, et al. Charged particle radiation therapy for uveal melanoma: A systematic review and meta-analysis. *Int J Radiat Oncol Biol Phys* 2013;86:18-26.
13. Mishra KK, Quivey JM, Daftari IK, et al. Long-term results of the UCSF-LBNL randomized trial: Charged particle with helium ion versus iodine-125 plaque therapy for choroidal and ciliary body melanoma. *Int J Radiat Oncol Biol Phys* 2015;92:376-383.
14. Bensoussan E, Thariat J, Maschi C, et al. Outcomes after proton beam therapy for large choroidal melanomas in 492 patients. *Am J Ophthalmol* 2016;165:78-87.
15. Zambrano AD, Chinela A, Bunge H, et al. Radiochirurgia estereotaxica de los melanomas uveales. Protocollo de tramiento. *Arch Oftalmol Buenos Aires* 1989;64:49-56.
16. Kang DW, Lee SC, Park YG, et al. Long-term results of Gamma Knife surgery for uveal melanomas. *J Neurosurg* 2012;117 Suppl:108-114.
17. Modorati G, Miserocchi E, Galli L, et al. Gamma knife radiosurgery for uveal melanoma: 12 years of experience. *Br J Ophthalmol* 2009; 93:40-44.
18. Fakiris AJ, Lo SS, Henderson MA, et al. Gamma-knife-based stereotactic radiosurgery for uveal melanoma. *Stereotact Funct Neurosurg* 2007;85:106-112.
19. Sarici AM, Pazarli H. Gamma-knife-based stereotactic radiosurgery for medium- and large-sized posterior uveal melanoma. *Graefes Arch Clin Exp Ophthalmol* 2013;251:285-294.
20. Mueller AJ, Talies S, Schaller UC, et al. Stereotactic radiosurgery of large uveal melanomas with the gamma-knife. *Ophthalmology* 2000; 107:1381-1387. discussion 1387-1388.
21. Dieckmann K, Georg D, Zehetmayer M, et al. LINAC based stereotactic radiotherapy of uveal melanoma: 4 years clinical experience. *Radiother Oncol* 2003;67:199-206.
22. Dunavoelgyi R, Dieckmann K, Gleiss A, et al. Local tumor control, visual acuity, and survival after hypofractionated stereotactic photon radiotherapy of choroidal melanoma in 212 patients treated between 1997 and 2007. *Int J Radiat Oncol Biol Phys* 2011;81:199-205.
23. Muller K, Nowak PJ, de Pan C, et al. Effectiveness of fractionated stereotactic radiotherapy for uveal melanoma. *Int J Radiat Oncol Biol Phys* 2005;63:116-122.
24. Muller K, Naus N, Nowak PJ, et al. Fractionated stereotactic radiotherapy for uveal melanoma, late clinical results. *Radiother Oncol* 2012;102:219-224.
25. Biewald E, Lautner H, Gok M, et al. Endoresection of large uveal melanomas: Clinical results in a consecutive series of 200 cases. *Br J Ophthalmol* 2017;1010:204-208.
26. Daftari IK, Petti PL, Shrieve DC, et al. Newer radiation modalities for choroidal tumors. *Int Ophthalmol Clin* 2006;46:69-79.
27. Zorlu F, Selek U, Kiratli H. Initial results of fractionated CyberKnife radiosurgery for uveal melanoma. *J Neurooncol* 2009;94:111-117.
28. Muacevic A, Nentwich M, Wowra B, et al. Development of a streamlined, non-invasive robotic radiosurgery method for treatment of uveal melanoma. *Technol Cancer Res Treat* 2008;7:369-374.
29. Choi SY, Kim MS, Yoo SY, et al. Feasibility of image-guided robotic radiotherapy using three fractions for uveal melanoma. *Tumori* 2009; 95:720-725.
30. Eibl-Lindner K, Furweger C, Nentwich M, et al. Robotic radiosurgery for the treatment of medium and large uveal melanoma. *Melanoma Res* 2016;26:51-57.
31. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma II: Initial mortality findings. COMS report no. 10. *Am J Ophthalmol* 1998;125:779-796.
32. Finger PT. Radiation therapy for choroidal melanoma. *Surv Ophthalmol* 1997;42:215-232.
33. Char DH, Kroll SM, Castro J. Ten-year follow-up of helium ion therapy for uveal melanoma. *Am J Ophthalmol* 1998;125:81-89.
34. Lodge M, Pijls-Johannesma M, Stirk L, et al. A systematic literature review of the clinical and cost-effectiveness of hadron therapy in cancer. *Radiother Oncol* 2007;83:110-122.
35. Wen JC, Oliver SC, McCannel TA. Ocular complications following I-125 brachytherapy for choroidal melanoma. *Eye (Lond)* 2009;23:1254-1268.
36. Klingenstein A, Furweger C, Muhlhofer AK, et al. Quality of life in the follow-up of uveal melanoma patients after enucleation in comparison to CyberKnife treatment. *Graefes Arch Clin Exp Ophthalmol* 2016;254:1005-1012.